

SECSR2043 OPERATING SYSTEMS

[20 Marks]

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 Section : 3

Marks

Instruction: Please answer all of the following questions. Whenever the 🖐 symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname.sh* (Example: ahmad.sh).

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*.c .; cp ../fruits/*.c .;
cp ../*.jar .
```

- a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **\$HOME** directory.

[4 marks]



Print screen the script that you type;

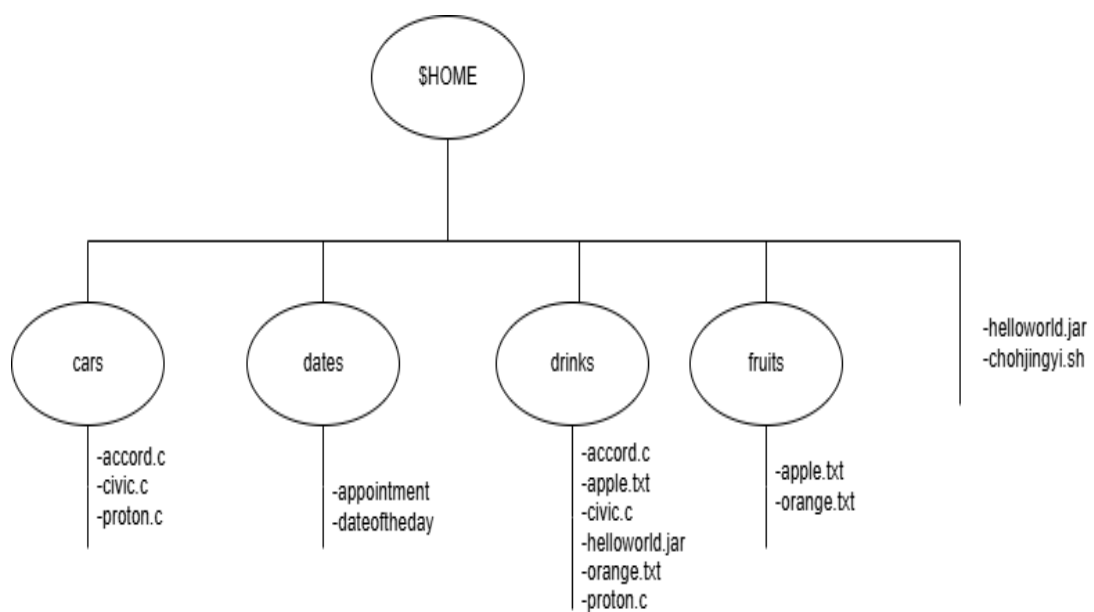
```

GNU nano 7.2 chohjingyi.sh
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt > orange.txt
cd ../drinks; cp ../cars/*.c .; cp ../fruits/*.c .;
cp ../*.jar .
```

Then draw the tree

```
chohjingyi@secl2043:~$ tree
```

```
.
├── cars
│   ├── accord.c
│   ├── civic.c
│   └── proton.c
├── chohjingyi.sh
├── dates
│   ├── appointment
│   └── dateoftheday
├── drinks
│   ├── accord.c
│   ├── apple.txt
│   ├── civic.c
│   ├── helloworld.jar
│   ├── orange.txt
│   └── proton.c
├── fruits
│   ├── apple.txt
│   └── orange.txt
└── helloworld.jar
```



- b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display c program files, and enter “c” as the input to the bash script. [4 marks]

Print screen the bash script you type and run

```
GNU nano 7.2 ifiles.sh
#!/bin/bash

#Prompt the user to enter a file extension
read -p "Enter the file extension (without the dot): " ext

#Find and display all files with the given extension
echo "Files with.${ext} extension:"
find . -type f -name "*.${ext}"

#Count the number of files with the given extension
count=$(find . -type f -name "*.${ext}" | wc -l)
echo "Number of files with .${ext} extension: $count"
```

```
chohjingyi@secr2043:~$ chmod u+x ifiles.sh
chohjingyi@secr2043:~$ ./ifiles.sh
Enter the file extension (without the dot): c
Files with.c extension:
./drinks/civic.c
./drinks/accord.c
./drinks/proton.c
./cars/civic.c
./cars/accord.c
./cars/proton.c
Number of files with .c extension: 6
chohjingyi@secr2043:~$
```

2. The following Figure 1 illustrates a tree structure of some directories and files.

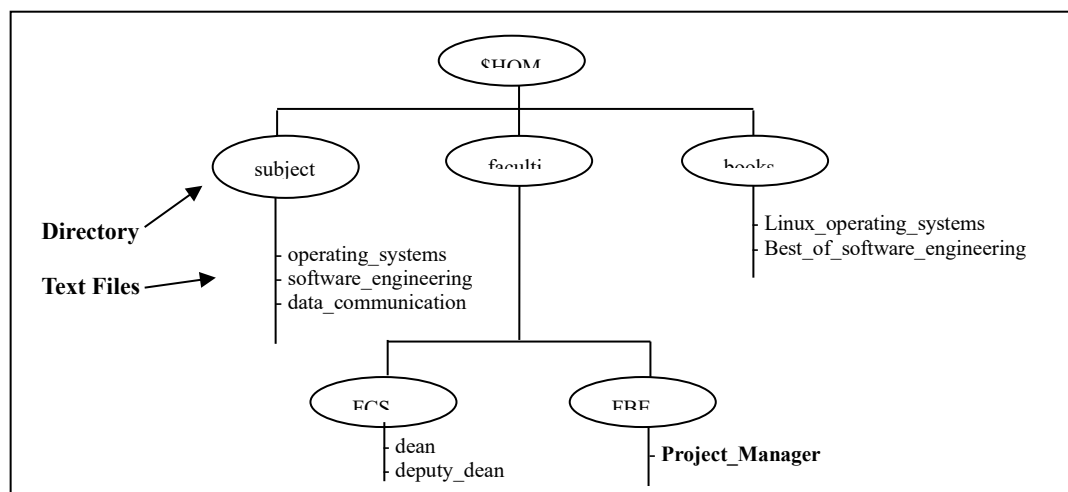
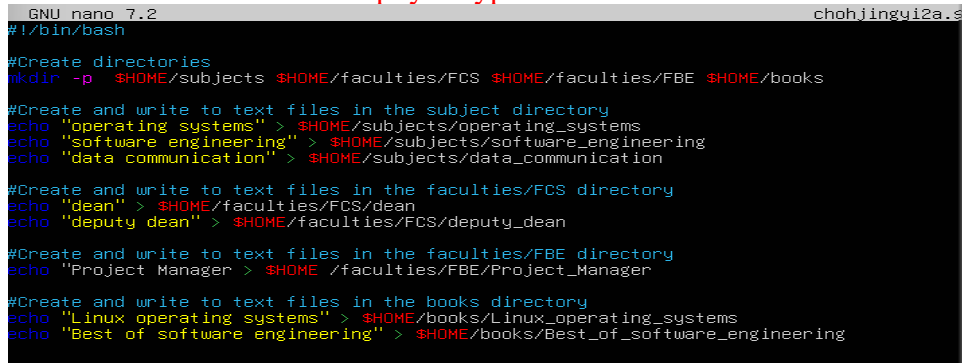


Figure 1

- a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

Print screen the bash script you type and run



```
GNU nano 7.2 chohjingyi2a.sh
#!/bin/bash

#Create directories
mkdir -p $HOME/subjects $HOME/faculties/FCS $HOME/faculties/FBE $HOME/books

#Create and write to text files in the subject directory
echo "operating systems" > $HOME/subjects/operating_systems
echo "software engineering" > $HOME/subjects/software_engineering
echo "data communication" > $HOME/subjects/data_communication

#Create and write to text files in the faculties/FCS directory
echo "dean" > $HOME/faculties/FCS/dean
echo "deputy dean" > $HOME/faculties/FCS/deputy_dean

#Create and write to text files in the faculties/FBE directory
echo "Project Manager" > $HOME/faculties/FBE/Project_Manager

#Create and write to text files in the books directory
echo "Linux operating systems" > $HOME/books/Linux_operating_systems
echo "Best of software engineering" > $HOME/books/Best_of_software_engineering
```

```
chohjingyi@secr2043:~$ chmod u+x chohjingyi2a.sh
chohjingyi@secr2043:~$ ./chohjingyi2a.sh
chohjingyi@secr2043:~$ tree $HOME
/home/chohjingyi
├── books
│   ├── Best_of_software_engineering
│   └── Linux_operating_systems
├── cars
│   ├── accord.c
│   ├── civic.c
│   └── proton.c
├── chohjingyi.sh
├── chohjingyi2a.sh
├── dates
│   ├── appointment
│   └── dateoftheday
├── drinks
│   ├── accord.c
│   ├── apple.txt
│   ├── civic.c
│   ├── helloworld.jar
│   ├── orange.txt
│   └── proton.c
├── faculties
│   ├── FBE
│   │   └── Project_Manager
│   └── FCS
│       ├── dean
│       └── deputy_dean
├── fruits
│   ├── apple.txt
│   └── orange.txt
├── helloworld.jar
├── ifiles.sh
└── subjects
    ├── data_communication
    ├── operating_systems
    └── software_engineering

10 directories, 25 files
chohjingyi@secr2043:~$
```



- b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called `book`.

[2 marks]

Directory/File	Access Control
books	drwxrwxr-x
subjects	drwxrwxr-x
Best_of_software_engineering	-rw-rw-r--
FCS	drwxrwxr-x
project_manager	-rw-rw-r--



```
drwxrwxr-x 2 chohjingyi chohjingyi 4096 Jun 13 14:58 books
```

```
drwxr--r-- 2 chohjingyi chohjingyi 4096 Jun 13 14:55 subjects
```

```
drwxrwxr-x 4 chohjingyi chohjingyi 4096 Jun 13 14:55 faculties
```

```
-r-x-w--- 1 chohjingyi chohjingyi 29 Jun 13 14:58 Best_of_software_engineering
-rw-rw-r-- 1 chohjingyi chohjingyi 24 Jun 13 15:28 Linux_operating_systems
subjects:
total 12
-rw-rw-r-- 1 chohjingyi chohjingyi 19 Jun 13 15:28 data_communication
-rw-rw-r-- 1 chohjingyi chohjingyi 18 Jun 13 15:28 operating_systems
-rw-rw-r-- 1 chohjingyi chohjingyi 21 Jun 13 15:28 software_engineering
chohjingyi@ees2013:~$
```

- c) Write another bash script (called `myname2c.sh`) that will change the access control of the directories and files based on the following information:

[4 marks]

Directory/File	Users								
	Owner			Group			Public		
subjects	✓	✓	✓	✓	x	x	✓	x	x
Best_of_software_engineering	✓	x	✓	x	✓	x	x	x	x
FCS	✓	✓	x	x	x	x	✓	✓	✓
project_manager	x	x	x	x	✓	✓	x	x	✓

Print screen the bash script you type and run

```
GNU nano 7.2 chohjingyi2c.s
#!/bin/bash

#Set permissions for subjects directory
chmod 744 $HOME/subjects

#Set permissions for Best_of_software_engineering file
chmod 520 $HOME/books/Best_of_software_engineering

#Set permissions for FCS directory
chmod 607 $HOME/faculties/FCS

#Set permissions for project_manager file
chmod 031 $HOME/faculties/FBE/Project_Manager
```

- d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]



Directory/File	Access Control
subjects	drwxr--r--
Best_of_software_engineering	-r-x-w----
FCS	drw----rwx
project_manager	----wx--x

```
drwxr--r-- 2 chohjingyi chohjingyi 4096 Jun 13 14:55 /home/chohjingyi/subjects
drw----rwx 2 chohjingyi chohjingyi 4096 Jun 13 14:55 /home/chohjingyi/faculties/FCS
-r-x-w---- 1 chohjingyi chohjingyi 29 Jun 13 14:58 Best_of_software_engineering
---wx--x 1 chohjingyi chohjingyi 16 Jun 13 14:58 /home/chohjingyi/faculties/FBE/Project_Manager
```

End of Lab 3

*** All the Best for Final Exam ***